GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Independent Set 4.01 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $(4.1\times10^{6})+(2.7\times10^{6})$

6.8×10^{6}

2. $(9.3\times10^{-5})-(5.1\times10^{-5})$

4.2×10^{-5}

3. $(7.5\times10^{3})+(8.9\times10^{3})$

1.64×10^{4}

The sum is 16.4×10^{3} *, which is not finished.*

4. $(6.2\times10^{7})+(3.5\times10^{6})$

6.55×10^{7}

$3.5\times10^{6}=0.35\times10^{7}$.

5. $(4.8\times10^{-2})-(9.0\times10^{-4})$

4.71×10^{-2}

$9.0\times10^{-4}=0.09\times10^{-2}$.

6. Find n . $(5.2\times10^{9})+(n\times10^{9})=1.31\times10^{10}$

***n* = 7.9**

$1.31\times10^{10}=13.1\times10^{9}$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Question Bank 4.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(2.4\times10^{5})+(3.1\times10^{5})$

5.5×10^{5}

2. $(8.3\times10^{-4})-(2.1\times10^{-4})$

6.2×10^{-4}

3. $(6.7\times10^{9})+(5.8\times10^{9})$

$12.5\times10^{9}=1.25\times10^{10}$

The sum of the coefficients is over 10, so the answer needs one more step.

4. $(9.1\times10^{4})-(8.6\times10^{4})$

$0.5\times10^{4}=5\times10^{3}$

A coefficient below 1 is unfinished too: point one place right, exponent down one.

5. $(7.2\times10^{6})+(4.3\times10^{5})$

7.63×10^{6}

$4.3\times10^{5}=0.43\times10^{6}$ *, and* $7.2+0.43=7.63$.

6. $(5.5\times10^{-3})-(8.0\times10^{-5})$

5.42×10^{-3}

$8.0\times10^{-5}=0.08\times10^{-3}$. *The exponents are two apart, the widest gap allowed.*

Real quantities.

7. Neptune's diameter is about 4.95×10^4 kilometers. Saturn's is about 1.21×10^5 kilometers.

a. What is the sum of the two diameters?

1.705×10^5 ***km***

$1.21 \times 10^5 = 12.1 \times 10^4$ *, so the sum is* 17.05×10^4 .

b. Jupiter's diameter is about 1.43×10^5 kilometers. Is your answer to part (a) greater than Jupiter's diameter?

Yes — $1.705 \times 10^5 > 1.43 \times 10^5$ **.**

Same power of ten, so the coefficients decide.

8. A city's population is about 4.62×10^5 . The surrounding county's population is about 1.38×10^6 .

a. What is the total population of the city and the county?

1.842×10^6 ***people***

$4.62 \times 10^5 = 0.462 \times 10^6$.

b. How many more people live in the county than in the city?

9.18×10^5 ***people***

$1.38 \times 10^6 = 13.8 \times 10^5$ *, and* $13.8 - 4.62 = 9.18$.

Working backwards.

9. Find the missing number in each equation.

a. $(3.6 \times 10^7) + (n \times 10^7) = 1.01 \times 10^8$

n = 6.5

$1.01 \times 10^8 = 10.1 \times 10^7$.

b. $(8.4 \times 10^{-6}) - (n \times 10^{-6}) = 2.9 \times 10^{-6}$

n = 5.5

c. $(2.5 \times 10^4) + (7.5 \times 10^4) =$ _____

1×10^5

The coefficients sum to exactly 10, so the answer is 10×10^4 *— finished, that is* 1×10^5 .

Reading someone else's work.

10. Priya writes $(2.6\times10^4)+(5.1\times10^6)=7.7\times10^{10}$. Name the two mistakes she made, and give the correct sum.

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 5.126×10^6 ***.***

$2.6\times10^4=0.026\times10^6$ *, and* $5.1+0.026=5.126$.

11. Which is the value of $(4.7\times10^{-5})+(8.6\times10^{-5})$, written in scientific notation?

- ○ A. 13.3×10^{-10}
- ● **B.** 1.33×10^{-4}
- ○ C. 1.33×10^{-10}
- ○ D. 13.3×10^{-5}

B

12. Select ALL of the expressions whose value is 1.2×10^5 .

- ☒ **A.** $(1.25\times10^5)-(5\times10^3)$
- ☐ B. $(1.3\times10^5)-(1\times10^5)$
- ☒ **C.** $(1.3\times10^5)-(1\times10^4)$
- ☐ D. $(8\times10^4)+(4\times10^5)$
- ☒ **E.** $(8\times10^4)+(4\times10^4)$
- ☒ **F.** $(1.1\times10^5)+(1\times10^4)$

A, C, E and F

E gives 12×10^4 *, which is* 1.2×10^5 *once it is finished.*

Adding and Subtracting in Scientific Notation

Question Bank - Additional 4.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(5.3\times10^{7})+(2.4\times10^{7})$

7.7×10^{7}

2. $(9.6\times10^{-3})-(4.2\times10^{-3})$

5.4×10^{-3}

3. $(7.4\times10^{5})+(8.9\times10^{5})$

$16.3\times10^{5}=1.63\times10^{6}$

4. $(6.3\times10^{-7})-(5.9\times10^{-7})$

$0.4\times10^{-7}=4\times10^{-8}$

Point one place right, exponent down one.

5. $(3.8\times10^{9})+(6.1\times10^{8})$

4.41×10^{9}

$6.1\times10^{8}=0.61\times10^{9}$.

6. $(2.7\times10^{-2})-(9.0\times10^{-4})$

2.61×10^{-2}

$9.0\times10^{-4}=0.09\times10^{-2}$.

Real quantities.

7. Mars's diameter is about 6.79×10^{3} kilometers. Earth's is about 1.28×10^{4} kilometers.

a. What is the sum of the two diameters?

1.959×10^{4} ***km***

$6.79\times10^{3}=0.679\times10^{4}$.

b. How much larger is Earth's diameter than Mars's?

6.01×10^{3} ***km***

$1.28\times10^{4}=12.8\times10^{3}$ *, and* $12.8-6.79=6.01$.

8. A charity raised about 4.85×10^5 dollars in its first year and about 1.27×10^6 dollars in its second.

a. How much did it raise in the two years together?

1.755×10^6 ***dollars***

$4.85 \times 10^5 = 0.485 \times 10^6$.

b. How much more did it raise in the second year than in the first?

7.85×10^5 ***dollars***

$1.27 \times 10^6 = 12.7 \times 10^5$.

Working backwards.

9. Find the missing number in each equation.

a. $(5.2 \times 10^8) + (n \times 10^8) = 1.34 \times 10^9$

n = 8.2

$1.34 \times 10^9 = 13.4 \times 10^8$.

b. $(9.6 \times 10^{-4}) - (n \times 10^{-4}) = 3.8 \times 10^{-4}$

n = 5.8

c. $(6.5 \times 10^3) + (3.5 \times 10^3) =$ _____

1×10^4

The coefficients sum to exactly 10.

Reading someone else's work.

10. Malik writes $(3.2 \times 10^5) + (7.1 \times 10^7) = 1.03 \times 10^{13}$. Name the two mistakes he made, and give the correct sum.

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 7.132×10^7 ***.***

$3.2 \times 10^5 = 0.032 \times 10^7$ *, and* $7.1 + 0.032 = 7.132$ *. His* 10.3×10^{12} *was then repaired correctly to* 1.03×10^{13} *— the repair is the one thing he did right.*

11. Which is the value of $(6.8 \times 10^{-6}) + (5.4 \times 10^{-6})$, written in scientific notation?

- ● **A.** 1.22×10^{-5}
- ○ B. 12.2×10^{-12}
- ○ C. 12.2×10^{-6}
- ○ D. 1.22×10^{-12}

A

12. Select ALL of the expressions whose value is 2.4×10^6 .

- ☒ **A.** $(1.2 \times 10^6) + (1.2 \times 10^6)$
- ☐ B. $(1.8 \times 10^6) + (6 \times 10^6)$
- ☒ **C.** $(2.42 \times 10^6) - (2 \times 10^4)$
- ☒ **D.** $(1.8 \times 10^6) + (6 \times 10^5)$
- ☒ **E.** $(2.5 \times 10^6) - (1 \times 10^5)$
- ☐ F. $(2.5 \times 10^6) - (1 \times 10^6)$

A, C, D and E

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Independent Set 4.02 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $(2\times10^5)(4\times10^3)$

8×10^8

2. $\frac{9.9\times10^7}{3.3\times10^2}$

3×10^5

3. $(5\times10^6)(6\times10^{-2})$

3×10^5

The product is 30×10^4 *, which is not finished.*

4. $\frac{1.8\times10^3}{6\times10^7}$

3×10^{-5}

The quotient is 0.3×10^{-4} *, which is not finished either.*

5. $\frac{4\times10^9}{8\times10^{-2}}$

5×10^{10}

$9-(-2)=11$ *, and* 0.5×10^{11} *is not finished.*

6. Find n . $(2\times10^n)(3\times10^4)=6\times10^{10}$

n = 6

$n+4=10$.

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Question Bank 4.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(2\times10^{3})(4\times10^{5})$

8×10^{8}

2. $\frac{8.8\times10^{9}}{2.2\times10^{4}}$

4×10^{5}

3. $(6\times10^{4})(5\times10^{7})$

$30\times10^{11}=3\times10^{12}$

The coefficients multiply to 30, so the answer needs one more step.

4. $\frac{1.5\times10^{6}}{5\times10^{9}}$

$0.3\times10^{-3}=3\times10^{-4}$

A coefficient below 1 is unfinished: point one place right, exponent down one.

5. $(3.5\times10^{-6})(4\times10^{-3})$

$14\times10^{-9}=1.4\times10^{-8}$

Both exponents are negative, so the product law adds them to −9.

6. $\frac{4.8\times10^{-2}}{1.6\times10^{-7}}$

3×10^{5}

$-2-(-7)=5$

Real quantities.

7. Three meteorites have masses 1.1×10^2, 6.8×10^2 and 8.4×10^{-2} grams.

a. How many times as heavy as the lightest one is the heaviest one? Round to the nearest whole number.

about 8,095 times

$680 \div 0.084 = 8{,}095.2\ldots$

b. How many times as heavy as the lightest one is the 1.1×10^2 gram one? Round to the nearest whole number.

about 1,310 times

$110 \div 0.084 = 1{,}309.5\ldots$

8. A data center processes about 4.5×10^6 records per hour.

a. How many records does it process in 2.4×10^2 hours?

1.08×10^9 ***records***

$4.5 \cdot 2.4 = 10.8$ *, so the product is* 10.8×10^8 *before it is finished.*

b. A second center processed 9.0×10^9 records in 1.8×10^3 hours. How many records per hour is that?

5×10^6 ***records per hour***

Working backwards.

9. Find the missing number in each equation.

a. $(3 \times 10^n)(2 \times 10^4) = 6 \times 10^{11}$

n = 7

$n + 4 = 11$.

b. $\frac{9.6 \times 10^8}{n \times 10^3} = 1.2 \times 10^5$

n = 8

$9.6 \div n = 1.2$.

c. $(5 \times 10^4)(4 \times 10^6) =$ _____

2×10^{11}

The coefficients multiply to 20, so 20×10^{10} *becomes* 2×10^{11} .

Reading someone else's work.

10. Tyree multiplies $(2.8\times10^3)(9.7\times10^6)$ and writes 27.16×10^9. Is his value correct? Is his answer written in scientific notation? Write the finished answer.

The value is correct. It is not in scientific notation, because 27.16 is not less than 10. Finished: 2.716×10^{10}.

$2.8\cdot9.7=27.16$ *and* $10^3\cdot10^6=10^9$ *, so every step he took was right — he stopped one step early.*

11. Which is the value of $\frac{7.5\times10^{-4}}{2.5\times10^2}$?

- ○ A. 5×10^{-6}
- ● **B.** 3×10^{-6}
- ○ C. 3×10^6
- ○ D. 3×10^{-2}

B

12. Select ALL of the expressions whose value is 6×10^8.

- ☒ **A.** $(1.5\times10^4)(4\times10^4)$
- ☒ **B.** $\frac{3\times10^{10}}{5\times10^1}$
- ☒ **C.** $(2\times10^3)(3\times10^5)$
- ☐ D. $\frac{1.2\times10^{12}}{2\times10^4}$
- ☐ E. $(2\times10^3)(3\times10^{15})$
- ☒ **F.** $\frac{1.2\times10^{12}}{2\times10^3}$

A, B, C and F

F and B each give 0.6×10^9 *, which is* 6×10^8 *once it is finished.*

Multiplying and Dividing in Scientific Notation

Question Bank - Additional 4.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(3\times10^{2})(3\times10^{6})$

9×10^{8}

2. $\frac{6.3\times10^{7}}{2.1\times10^{3}}$

3×10^{4}

3. $(4\times10^{5})(8\times10^{6})$

$32\times10^{11}=3.2\times10^{12}$

4. $\frac{2.4\times10^{4}}{8\times10^{7}}$

$0.3\times10^{-3}=3\times10^{-4}$

5. $(2.5\times10^{-5})(6\times10^{-4})$

$15\times10^{-9}=1.5\times10^{-8}$

6. $\frac{9.9\times10^{-3}}{3.3\times10^{-8}}$

3×10^{5}

$-3-(-8)=5$.

Real quantities.

7. The Amazon River discharges about 5.5×10^{7} gallons of water every second.

a. About how many gallons does it discharge in one hour, which is 3.6×10^{3} seconds?

about 1.98×10^{11} ***gallons***

$5.5\cdot3.6=19.8$ *, so the product is* 19.8×10^{10} *before it is finished.*

b. A second river discharges about 1.1×10^{7} gallons per second. How many times larger is the Amazon's rate?

5 times larger

8. A factory produces about 6.4×10^5 bottles per day.

a. How many bottles does it produce in 3.5×10^2 days?

2.24×10^8 ***bottles***

$6.4 \cdot 3.5 = 22.4$.

b. A second factory produced 7.2×10^8 bottles in 1.2×10^3 days. How many bottles per day is that?

6×10^5 ***bottles per day***

Working backwards.

9. Find the missing number in each equation.

a. $(4 \times 10^n)(2 \times 10^3) = 8 \times 10^{10}$

n = 7

$n + 3 = 10$.

b. $\frac{7.2 \times 10^6}{n \times 10^2} = 2.4 \times 10^4$

n = 3

$7.2 \div n = 2.4$.

c. $(2.5 \times 10^3)(8 \times 10^5) =$ _____

2×10^9

The coefficients multiply to exactly 20.

Reading someone else's work.

10. Dana divides $\frac{4.8 \times 10^{11}}{1.2 \times 10^4}$ and writes 3.6×10^7 . Name her mistake and give the correct quotient.

She subtracted the coefficients instead of dividing them. The power of ten is right. The correct quotient is 4×10^7 ***.***

$4.8 - 1.2 = 3.6$ *, which is where her coefficient came from;* $4.8 \div 1.2 = 4$.

11. Which is the value of $\frac{9.6 \times 10^{-3}}{3.2 \times 10^5}$?

- ● **A.** 3×10^{-8}
- ○ B. 3×10^8
- ○ C. 6.4×10^{-8}
- ○ D. 3×10^2

A

12. Select ALL of the expressions whose value is 8×10^6 .

- ☒ **A.** $(1.6 \times 10^3)(5 \times 10^3)$
- ☒ **B.** $\frac{1.6 \times 10^{10}}{2 \times 10^3}$
- ☐ C. $(4 \times 10^2)(2 \times 10^8)$
- ☒ **D.** $\frac{4 \times 10^8}{5 \times 10^1}$
- ☒ **E.** $(4 \times 10^2)(2 \times 10^4)$
- ☐ F. $\frac{1.6 \times 10^{10}}{2 \times 10^4}$

A, B, D and E

B and D each give 0.8×10^7 *, which is* 8×10^6 *once it is finished.*

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Independent Set 4.04 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. How many significant digits are in 6,030 grams?

3

The middle zero counts; the trailing zero does not, because no decimal point is written.

2. How many significant digits are in 0.00710 meters?

3

Leading zeros never count; the trailing zero does, because a decimal point is written.

3. A rectangle measures 2.4×10^2 m by 3.15×10^2 m. Find its area, to the correct number of significant digits.

7.6×10^4 ***square meters***

75,600 exactly, kept to two significant digits.

4. $(6.30 \times 10^4) + (2.8 \times 10^3)$ meters, both measured. Write the sum with the correct precision.

6.58×10^4 ***m***

63,000 and 2,800 both stop at the hundreds place, so the sum, 65,800, does too.

5. A car travels 1.80×10^2 km in 4.0 hours. Find its average speed, to the correct number of significant digits.

4.5×10^1 ***km/h***

45 exactly; two significant digits.

6. $(3.7 \times 10^{-2}) - (8.45 \times 10^{-4})$ grams, both measured. Write the difference with the correct precision.

3.6×10^{-2} ***g***

0.036155 exactly; 0.037 stops at the thousandths place, so the answer stops there.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Question Bank 4.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, give the number of significant digits in each measurement.

1. 4,207 grams

4

Every digit counts: the zero sits between two nonzero digits.

2. 0.0058 meters

2

Leading zeros are never significant.

3. 3.200 seconds

4

The trailing zeros count, because a decimal point is written. 3.200 and 3.2 are the same number but not the same claim.

4. 52,000 litres

2

No decimal point is written, so the trailing zeros are only holding places.

For questions 5 through 8, solve. Report each answer with the correct precision.

5. A rectangular plot measures 3.5×10^2 m by 1.24×10^2 m. Find its area.

4.3×10^4 ***square meters***

43,400 exactly; 2 significant digits and 3, so the answer keeps 2.

6. A car travels 2.60×10^2 km in 3.0 hours. Find its average speed.

8.7×10^1 ***km/h***

260 ÷ 3.0 = 86.66…; 3 significant digits and 2, so the answer keeps 2.

7. Two lengths are measured: 8.7×10^2 mm and 4.5×10^2 mm. Find their sum.

1.32×10^3 ***mm***

870 and 450 both stop at the tens place, so the sum, 1,320, does too. A sum can carry more significant digits than either measurement — three here, from two twos.

8. Two masses are measured: 8.40×10^{-2} g and 3.715×10^{-3} g. Find their difference.

8.03×10^{-2} ***g***

0.080285 exactly; 0.0840 stops at the ten-thousandths place and 0.003715 stops two places further out, so the answer stops at the ten-thousandths.

Real quantities.

9. The Amazon River discharges about 5.5×10^{7} gallons of water each second.

a. How many gallons does it discharge in one hour, which is 3.6×10^{3} seconds? Report the answer with the correct number of significant digits.

2.0×10^{11} ***gallons***

1.98×10^{11} *exactly. The rate carries two significant digits; 3,600 seconds in an hour is an exact conversion, not a measurement, so it does not limit the count. The answer keeps two.*

b. A second river discharges about 1.10×10^{7} gallons per second. How many times larger is the Amazon's rate? Report the answer with the correct number of significant digits.

5.0 times larger

Exactly 5; two significant digits are justified, so it is written 5.0.

10. A processor performs about 4.66×10^{8} calculations each second.

a. How many calculations does it perform in one minute? Report the answer with the correct number of significant digits.

2.80×10^{10} ***calculations***

$4.66 \times 10^{8} \cdot 60 = 2.796 \times 10^{10}$ *. Sixty seconds in a minute is exact, not a measurement, so it does not limit the count — the answer keeps the measurement's three digits.*

b. Five of these processors run for 20 minutes. How many calculations do they perform altogether? Report the answer with the correct number of significant digits.

2.80×10^{12} ***calculations***

$2.796 \times 10^{10} \cdot 5 \cdot 20 = 2.796 \times 10^{12}$ *. Five and twenty are counts, so again only the measurement's three digits survive.*

Reading someone else's work.

11. Omar multiplies a measured length of 2.5×10^{2} m by a measured width of 1.346×10^{2} m and reports the area as 3.365×10^{4} square meters. Is his arithmetic correct? Is his answer correct? Explain.

His arithmetic is correct — the exact product is 3.365×10^{4} ***. His answer is not, because 2.5 carries only two significant digits, so the area can carry only two:*** 3.4×10^{4} ***square meters.***

A correct value in the wrong form is still the wrong answer here; the rule is part of the question.

12. A rectangle's sides are measured as 7.2×10^1 cm and 4.0×10^1 cm. Which is the area, in scientific notation with the correct number of significant digits?

- ○ A. 2.9×10^2 cm^2
- ○ B. 28.8×10^2 cm^2
- ● **C.** 2.9×10^3 **cm^2**
- ○ D. 2.88×10^3 cm^2

C

Real-World Problems and Significant Digits

Question Bank - Additional 4.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, give the number of significant digits in each measurement.

1. 9,304 kilograms

4

The zero sits between two nonzero digits.

2. 0.00062 meters

2

Leading zeros are never significant.

3. 1.500 seconds

4

The trailing zeros count because a decimal point is written.

4. 430,000 litres

2

No decimal point, so the trailing zeros only hold places.

For questions 5 through 8, solve. Report each answer with the correct precision.

5. A rectangular field measures 6.5×10^2 m by 2.34×10^2 m. Find its area.

1.5×10^5 ***square meters***

152,100 exactly; the answer keeps 2 significant digits.

6. A train travels 3.40×10^2 km in 4.0 hours. Find its average speed.

8.5×10^1 ***km/h***

340 ÷ 4.0 = 85 exactly; the answer keeps 2 significant digits.

7. Two lengths are measured: 6.28×10^4 mm and 9.1×10^3 mm. Find their sum.

7.19×10^4 ***mm***

62,800 stops at the hundreds place and 9,100 stops at the hundreds place, so the sum, 71,900, stops there too.

8. Two masses are measured: 5.70×10^{-1} g and 2.845×10^{-2} g. Find their difference.

5.42×10^{-1} ***g***

0.54155 exactly; 0.570 stops at the thousandths place, so the answer stops there.

Real quantities.

9. A wind turbine generates about 2.4×10^{3} kilowatt-hours of electricity each hour.

a. How much does it generate in 7.5×10^{3} hours? Report the answer with the correct number of significant digits.

1.8×10^{7} ***kilowatt-hours***

$18 \times 10^{6} = 1.8 \times 10^{7}$ *exactly; both measurements carry two significant digits.*

b. A second turbine generates about 8.0×10^{2} kilowatt-hours each hour. How many times larger is the first turbine's rate? Report the answer with the correct number of significant digits.

3.0 times larger

Exactly 3; two significant digits are justified, so it is written 3.0.

10. A sensor records about 8.25×10^{5} readings each second.

a. How many readings does it record in one minute? Report the answer with the correct number of significant digits.

4.95×10^{7} ***readings***

$8.25 \times 10^{5} \cdot 60 = 4.95 \times 10^{7}$ *exactly. Sixty is a count, not a measurement, so the three digits survive.*

b. Four of these sensors run for 30 minutes. How many readings do they record altogether? Report the answer with the correct number of significant digits.

5.94×10^{9} ***readings***

$4.95 \times 10^{7} \cdot 4 \cdot 30 = 5.94 \times 10^{9}$ *exactly. Four and thirty are counts.*

Reading someone else's work.

11. Bianca adds two measured lengths, 86.4 cm and 0.271 cm. She says that because 86.4 has three significant digits and 0.271 has three, the sum has three: 86.7 cm. Is she right? Explain, and give the correct sum.

She is not right, even though her answer happens to be close. The rule for a sum is about places, not counts: 86.4 stops at the tenths and 0.271 stops at the thousandths, so the sum stops at the tenths. $86.4 + 0.271 = 86.671$ ***, which becomes 86.7 cm.***

Here the two rules give the same answer, which is exactly why the wrong rule survives so long. Ask what she would have written for 1,234 + 5.6.

12. A rectangle's sides are measured as 5.0×10^1 cm and 6.4×10^1 cm. Which is the area, in scientific notation with the correct number of significant digits?

- ○ A. 3.2×10^2 cm^2
- ○ B. 3.20×10^3 cm^2
- ○ C. 32×10^2 cm^2
- ● **D.** 3.2×10^3 **cm^2**

D

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Independent Set 4.05 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $\sqrt{225}$

15

2. $\sqrt[3]{-1}$

−1

An odd root keeps the sign.

3. $\sqrt{6^2+13}$

7

36 + 13 = 49.

4. $\sqrt[3]{\frac{1}{8}}$

$\frac{1}{2}$

5. $\sqrt{144}\cdot 2^{-3}$

$\frac{3}{2}$

$12\cdot\frac{1}{8}$.

6. $\left(-\frac{1}{2}\right)^3+\sqrt{4^2+9}$

$\frac{39}{8}$

$-\frac{1}{8}+5$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Question Bank 4.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{81}$

9

2. $\sqrt[3]{125}$

5

3. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

4. $\sqrt{\frac{4}{25}}$

$\frac{2}{5}$

The numerator and the denominator each get the root.

5. $\sqrt{5^2+11}$

6

Finish under the bar: 25 + 11 = 36.

6. $\sqrt[3]{3^3+37}$

4

27 + 37 = 64.

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{100} \cdot 2^{-2}$

$\frac{5}{2}$

$10 \cdot \frac{1}{4}$.

b. $\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

c. $\sqrt[3]{-64} - 3^{-1}$

$-\frac{13}{3}$

$-4 - \frac{1}{3}$.

8. Evaluate each expression.

a. $\sqrt[3]{27} - 1.4\left(\sqrt{3^2 - 5}\right)$

0.2

$3 - 1.4(2)$.

b. $\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$

$-\frac{17}{9}$

$\frac{1}{9} - 2$.

c. $\sqrt{4^3 - 39} \cdot 5^{-3}$

$\frac{1}{25}$

$5 \cdot \frac{1}{125}$.

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 12$

n = 144

b. $\sqrt[3]{n} = -5$

n = −125

c. $\sqrt{n+7} = 9$

n = 74

$n + 7 = 81$.

Reading someone else's work.

10. Tonya writes $\sqrt{16+9} = \sqrt{16} + \sqrt{9} = 7$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that her method ignores.

$\sqrt{16+9} = \sqrt{25} = 5$ ***. The bar groups 16 + 9 into a single radicand, so the addition is finished before any root is taken.***

Her method would need the root of a sum to equal the sum of the roots, and it does not.

11. Which is the value of $\sqrt{36+64}$?

- ○ A. 14
- ○ B. 100
- ● **C. 10**
- ○ D. 50

C

12. Select ALL of the expressions whose value is 3.

- ☐ A. $\sqrt{3^2+16}$
- ☒ **B.** $\sqrt[3]{2^2+23}$
- ☒ **C.** $\sqrt{4^2-7}$
- ☒ **D.** $\sqrt[3]{27}$
- ☐ E. $\sqrt[3]{-27}$
- ☒ **F.** $\sqrt{9}$

B, C, D and F

C's radicand is $16-7=9$ *and B's is* $4+23=27$.

Evaluating Expressions with Radicals

Question Bank - Additional 4.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{64}$

8

2. $\sqrt[3]{8}$

2

3. $\sqrt[3]{-125}$

−5

An odd root keeps the sign of its radicand.

4. $\sqrt{\frac{9}{49}}$

$\frac{3}{7}$

Top and bottom each get the root.

5. $\sqrt{4^2+33}$

7

16 + 33 = 49.

6. $\sqrt[3]{5^2+100}$

5

25 + 100 = 125.

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{144} \cdot 3^{-2}$

$\frac{4}{3}$

$12 \cdot \frac{1}{9}$.

b. $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

$\frac{97}{16}$

$\frac{1}{16} + 6$.

c. $\sqrt[3]{-27} - 2^{-1}$

$-\frac{7}{2}$

$-3 - \frac{1}{2}$.

8. Evaluate each expression.

a. $\sqrt[3]{64} - 1.5\left(\sqrt{7^2 - 13}\right)$

−5

$4 - 1.5(6)$.

b. $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

$-\frac{25}{8}$

$-\frac{1}{8} - 3$.

c. $\sqrt{6^2 + 13} \cdot 2^{-3}$

$\frac{7}{8}$

$7 \cdot \frac{1}{8}$.

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 15$

n = 225

b. $\sqrt[3]{n} = -2$

n = −8

c. $\sqrt{n-5} = 11$

n = 126

$n - 5 = 121$.

Reading someone else's work.

10. Deshawn writes $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that his method ignores.

$\sqrt{100-36} = \sqrt{64} = 8$ ***. The bar groups the subtraction into a single radicand, so it is finished before any root is taken; the root of a difference is not the difference of the roots.***

His answer is half the true value. Ask him to try $\sqrt{25-9}$ *: his method gives 2 and the value is 4.*

11. Which is the value of $\sqrt{8^2+36}$?

- ○ A. 14
- ● **B. 10**
- ○ C. 100
- ○ D. 50

B

12. Select ALL of the expressions whose value is 4.

- ☐ A. $\sqrt{2^3+17}$
- ☒ **B.** $\sqrt[3]{64}$
- ☐ C. $\sqrt[3]{-64}$
- ☒ **D.** $\sqrt{16}$
- ☒ **E.** $\sqrt{5^2-9}$
- ☒ **F.** $\sqrt[3]{3^3+37}$

B, D, E and F

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Independent Set 4.06 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $30 - 4 \cdot 5 + 2$

12

2. $36 \div 6 \cdot 2$

12

One level, left to right.

3. $3 + 5(6 - 2)^2$

83

$3 + 5(16)$.

4. $\frac{5^2 - 1}{2^2 + 2}$

4

24 over 6.

5. $-4^2 + (-4)^2$

0

6. $\sqrt{100 - 19} + 2^{-2}$

$\frac{37}{4}$

Under the bar: 81, whose root is 9. Then $9 + \frac{1}{4}$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Question Bank 4.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $20 - 3 \cdot 4 + 2$

10

Multiply first, then left to right: 20 − 12 + 2.

2. $24 \div 4 \cdot 3$

18

One level, left to right: 6 · 3.

3. $5 + 2(7 - 3)^2$

37

Parentheses, then the power, then the product: 5 + 2(16).

4. $\frac{4^2 + 8}{2^3 - 5}$

8

24 over 3.

5. $-3^2 + (-3)^2$

0

The first exponent is attached to the 3 alone; the second is attached to −3.

6. $\sqrt{100 - 36} + 2^{-2}$

$\frac{33}{4}$

Under the bar first: 64, whose root is 8. Then $8 + \frac{1}{4}$.

For questions 7 and 8, evaluate. Each expression has more than one grouping symbol.

7. Evaluate each expression.

a. $4(8.63-(3.2)^2)+\sqrt[3]{125}$

−1.44

$4(-1.61)+5$.

b. $\frac{2^3-5}{6^2}-\sqrt{\frac{4}{9}-\frac{1}{3}}$

$-\frac{1}{4}$

$\frac{1}{12}-\frac{1}{3}$.

c. $12\left(\left(\frac{1}{2}\right)^3+3^{-2}\right)$

$\frac{17}{6}$

$12\cdot\frac{17}{72}$. *Written as* $\frac{51}{18}$ *it is the same number.*

8. Evaluate each expression.

a. $[3+(2-5)^2]\cdot 2^3$

96

Innermost first: $(-3)^2=9$ *, so the bracket is 12.*

b. $\frac{(-4)^3}{2^4}-\sqrt{81}$

−13

$-4-9$.

c. $\sqrt[3]{-27}\cdot\left(\frac{1}{3}\right)^{-2}$

−27

$-3\cdot 9$.

Working backwards.

9. Find the value of n that makes each statement true.

a. $n+3\cdot 4=20$

n = 8

b. $(n+1)^2=49$, where n is positive

n = 6

c. $\frac{n-4}{2^2}=5$

n = 24

$n-4=20$.

Reading someone else's work.

10. Deshawn says that $12 \div 3 \cdot 2 = 2$, because you always multiply before you divide. Is he right? Explain, and give the correct value.

He is not right. Multiplication and division are one level and are taken left to right, so $12 \div 3 = 4$ ***and then*** $4 \cdot 2 = 8$ ***.***

There is no rule that puts multiplication before division; the two share a level, and so do addition and subtraction.

11. Which is the value of $18 - 2(3)^2$?

- ○ A. 12
- ○ B. 144
- ○ C. −18
- ● **D. 0**

D

12. Select ALL of the expressions whose value is 4.

- ☐ A. -2^2
- ☒ **B.** $\left(\frac{1}{2}\right)^{-2}$
- ☒ **C.** $2 + 3 - 1$
- ☒ **D.** $\sqrt{16}$
- ☒ **E.** $\frac{6^2 - 4}{2^3}$
- ☐ F. $12 \div 3 \cdot 2$

B, C, D and E

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Question Bank - Additional 4.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $30 - 4 \cdot 6 + 5$

11

30 − 24 + 5.

2. $48 \div 8 \cdot 2$

12

One level, left to right: 6 · 2.

3. $7 + 3(9 - 5)^2$

55

7 + 3(16).

4. $\frac{5^2 + 11}{3^2 - 3}$

6

36 over 6.

5. $-5^2 + (-5)^2$

0

The exponent attaches to its own base; the parentheses decide what that base is.

6. $\sqrt{169 - 25} + 3^{-1}$

$\frac{37}{3}$

Under the bar: 144, whose root is 12.

For questions 7 and 8, evaluate. Each expression has more than one grouping symbol.

7. Evaluate each expression.

a. $5(7.42-(2.6)^2)+\sqrt[3]{64}$

7.3

$5(0.66)+4$.

b. $\frac{3^3-3}{4^2}-\sqrt{\frac{5}{9}-\frac{4}{9}}$

$\frac{7}{6}$

$\frac{3}{2}-\frac{1}{3}$.

c. $18\left(\left(\frac{1}{3}\right)^2+2^{-1}\right)$

11

$18\cdot\frac{11}{18}$.

8. Evaluate each expression.

a. $[5+(3-7)^2]\cdot 2^2$

84

Innermost first: $(-4)^2=16$ *, so the bracket is 21.*

b. $\frac{(-5)^3}{5^2}-\sqrt{121}$

−16

$-5-11$.

c. $\sqrt[3]{-8}\cdot\left(\frac{1}{2}\right)^{-3}$

−16

$-2\cdot 8$.

Working backwards.

9. Find the value of n that makes each statement true.

a. $n+5\cdot 3=27$

n = 12

b. $(n-2)^2=64$, where n is greater than 2

n = 10

c. $\frac{n+6}{3^2}=4$

n = 30

$n+6=36$.

Reading someone else's work.

10. Liesel says that $10-4+3=3$, because addition comes before subtraction. Is she right? Explain, and give the correct value.

She is not right. Addition and subtraction are one level and are taken left to right, so $10-4=6$ ***and then*** $6+3=9$ **.**

The same answer as Deshawn's, one level down: no rule puts either of a pair ahead of the other.

11. Which is the value of $25-3(2)^3$?

- ○ A. 7
- ○ B. −191
- ● **C. 1**
- ○ D. 6,859

C

12. Select ALL of the expressions whose value is 9.

- ☒ **A.** $\left(\frac{1}{3}\right)^{-2}$
- ☐ B. $36\div 2\cdot 2$
- ☒ **C.** $4+6-1$
- ☒ **D.** $\sqrt{81}$
- ☐ E. -3^2
- ☒ **F.** $\frac{7^2-4}{5}$

A, C, D and F

Real-World Order of Operations

Independent Set 4.07 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $6x^2$ when $x=-2$

24

$6(-2)^2=6(4)$.

2. $\frac{1}{2}mv^2$ when $m=90$ and $v=4$

720

$45(16)$.

3. $\frac{a^2-b}{c}$ when $a=9$, $b=6$, $c=5$

15

Finish the numerator first: $81-6=75$.

4. A square room of side 10 feet is carpeted at 2.65 dollars a square foot, with a flat 55 dollars for fitting. Find the total cost.

320 dollars

$100(2.65)+55$.

5. $2(x+y)^2-3$ when $x=4$ and $y=1$

47

$2(25)-3$.

6. A gym charges 25 dollars to join plus 8 dollars for each of 6 classes. Find the total cost.

73 dollars

$25+8(6)$ *— only the class price is repeated.*

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Real-World Order of Operations

Question Bank 4.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $4x^2$ when $x=-3$

36

$4(-3)^2=4(9)$.

2. $\frac{1}{2}mv^2$ when $m=50$ and $v=6$

900

$25(36)$.

3. $\frac{a+b^2}{c}$ when $a=6$, $b=5$, $c=2$

$\frac{31}{2}$

The bar groups the whole numerator: $6+25=31$.

4. $2(x-y)^2$ when $x=3$ and $y=7$

32

$2(-4)^2=2(16)$.

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

$11^2(3.20)+75=462.20$ ***dollars***

The rate is per square foot, so it multiplies the area; the delivery is flat, so it does not.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

$14(9)(1.15)+48=192.90$ ***dollars***

The area is 126 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 150$, $h = 65$

25.0

$\frac{105{,}450}{4{,}225} = 24.958\ldots$

b. $w = 180$, $h = 70$

25.8

$\frac{126{,}540}{4{,}900} = 25.824\ldots$

c. $w = 200$, $h = 72$

27.1

$\frac{140{,}600}{5{,}184} = 27.122\ldots$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1{,}200$, $v = 15$

135,000 joules

b. $m = 850$, $v = 12$

61,200 joules

c. $m = 70$, $v = 8$

2,240 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 900$ and $v = 6$. Find m.

m = 50

$\frac{1}{2}m(36) = 900$, *so* $18m = 900$.

b. $x^2 + 5 = 41$, where x is positive. Find x.

x = 6

c. A square lawn of side n feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find n.

n = 30

$n^2(2.50) = 2{,}250$, *so* $n^2 = 900$.

Reading the situation.

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.

A is $20+3(5)=35$ ***dollars. B is*** $20(3)+5=65$ ***dollars. Both problems contain the same numbers and the same two words, so the words cannot be what decides; what decides is which amount is repeated.***

This is the state guide's named misconception, tested directly: circling the numbers and boxing the key words produces identical markings on two problems with different answers.

11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?

- ○ A. $15(4.80)+120$
- ● **B.** $15^2(4.80)+120$
- ○ C. $15^2(4.80+120)$
- ○ D. $15^2+4.80+120$

B

12. Select ALL of the expressions whose value is 45 when $x=3$.

- ☐ A. $(5x)^2$
- ☒ **B.** $\frac{135}{x}$
- ☒ **C.** $5x^2$
- ☒ **D.** $3x^2+18$
- ☐ E. $5x\cdot 2$
- ☒ **F.** x^2+36

B, C, D and F

Real-World Order of Operations

Question Bank - Additional 4.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $5x^2$ when $x = -4$

80

$5(-4)^2 = 5(16)$.

2. $\frac{1}{2}mv^2$ when $m = 30$ and $v = 10$

1,500

$15(100)$.

3. $\frac{a^2 + b}{c}$ when $a = 8$, $b = 4$, $c = 17$

4

The bar groups the whole numerator: $64 + 4 = 68$.

4. $3(x + y)^2$ when $x = -5$ and $y = 2$

27

$3(-3)^2 = 3(9)$.

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

$13^2(2.75) + 90 = 554.75$ ***dollars***

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

$16(7)(1.40) + 35 = 191.80$ ***dollars***

The area is 112 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 120$, $h = 62$

21.9

$\frac{84,360}{3,844} = 21.945\ldots$

b. $w = 165$, $h = 68$

25.1

$\frac{115,995}{4,624} = 25.085\ldots$

c. $w = 210$, $h = 74$

27.0

$\frac{147,630}{5,476} = 26.959\ldots$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1,500$, $v = 20$

300,000 joules

b. $m = 640$, $v = 9$

25,920 joules

c. $m = 55$, $v = 4$

440 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 1,600$ and $v = 8$. Find m.

m = 50

$32m = 1,600$.

b. $x^2 - 7 = 74$, where x is positive. Find x.

x = 9

c. A square lawn of side n feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find n.

n = 20

$n^2(3.00) = 1,200$, *so* $n^2 = 400$.

Reading the situation.

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.

A is $8+12(4)=56$ ***dollars. B is*** $8(12)+4=100$ ***dollars. The two problems carry the same three numbers, so the numbers cannot be what decides; what decides is which amount is repeated and how often.***

Ask which number is a rate in each problem — that question has a different answer in A and in B, and no marking on the page can show it.

11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?

- ○ A. $20^2(3.60+200)$
- ○ B. $20(3.60)+200$
- ● **C.** $20^2(3.60)+200$
- ○ D. $20^2+3.60+200$

C

12. Select ALL of the expressions whose value is 32 when $x=4$.

- ☒ **A.** x^2+16
- ☒ **B.** $\frac{128}{x}$
- ☐ C. $2x\cdot 2$
- ☒ **D.** $x^2\cdot 2$
- ☐ E. $(2x)^2$
- ☒ **F.** $2x^2$

A, B, D and F

GRADE 7 ACCELERATED · UNIT 4

Unit 4 Review

Solving Problems with Rational Numbers **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers each part of the assessment, with more questions than the assessment has. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Adding and Subtracting in Scientific Notation

Lesson 1 · MA.8.NSO.1.5 · 6 questions

1. $(4.1\times10^{6})+(2.3\times10^{6})$

6.4×10^{6}

Same power: add the coefficients, copy the power down.

2. $(7.6\times10^{-4})-(2.9\times10^{-4})$

4.7×10^{-4}

3. $(8.2\times10^{5})+(4.9\times10^{5})$

1.31×10^{6}

13.1×10^{5} *is the value and not the finished form.*

4. $(5.4\times10^{8})+(7.1\times10^{7})$

6.11×10^{8}

$7.1\times10^{7}=0.71\times10^{8}$.

5. $(6.3\times10^{-2})-(8\times10^{-4})$

6.22×10^{-2}

$8\times10^{-4}=0.08\times10^{-2}$ *; the exponents are two apart, the widest gap allowed.*

6. Lena writes $(3.2\times10^{4})+(5.5\times10^{6})=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

She added the coefficients without first writing both over the same power of ten, and she added the exponents. The sum is 5.532×10^{6} **.**

$3.2\times10^{4}=0.032\times10^{6}$ *, and* $5.5+0.032=5.532$.

PART B. Multiplying and Dividing in Scientific Notation

Lessons 2–3 · MA.8.NSO.1.5 · 6 questions

7. $(4\times10^{3})(2\times10^{6})$

8×10^{9}

Multiply the coefficients, add the exponents.

8. $\frac{7.2\times10^{8}}{1.8\times10^{3}}$

4×10^{5}

Divide the coefficients, subtract the exponents, top minus bottom.

9. $(5\times10^{6})(8\times10^{-2})$

4×10^{5}

40×10^{4} *is not finished.*

10. $\frac{2.4\times10^{5}}{8\times10^{9}}$

3×10^{-5}

0.3×10^{-4} *is not finished either.*

11. $\frac{4\times10^{7}}{4\times10^{-3}}$

1×10^{10}

$7-(-3)=10$ *. Subtracting a negative goes up.*

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4×10^{2} seconds?

2.6×10^{17} ***photons***

$6.5\cdot4=26$ *, so* 26×10^{16} *before it is finished.*

PART C. Real-World Problems in Scientific Notation

Lesson 4 · MA.8.NSO.1.6 · 4 questions

13. A rectangular panel measures 3.2×10^{2} mm by 1.45×10^{2} mm. Find its area, in scientific notation. Do not round.

4.64×10^{4} ***square millimeters***

$3.2\cdot1.45=4.64$ *, and* $10^{2}\cdot10^{2}=10^{4}$ *.*

14. A train travels 4.8×10^{2} km in 6 hours. Find its average speed, in scientific notation. Do not round.

8×10^{1} ***km/h***

$\frac{4.8\times10^{2}}{6}=0.8\times10^{2}$ *, which is not finished.*

15. Two lengths are measured: 5.4×10^{3} mm and 2.8×10^{2} mm. Find their sum, in scientific notation. Do not round.

5.68×10^{3} ***mm***

$2.8\times10^{2}=0.28\times10^{3}$ *, and* $5.4+0.28=5.68$.

16. Two masses are measured: 7.6×10^{-2} g and 3.42×10^{-3} g. Find their difference, in scientific notation. Do not round.

7.258×10^{-2} ***g***

$3.42\times10^{-3}=0.342\times10^{-2}$ *, and* $7.6-0.342=7.258$.

PART D. Evaluating Expressions with Radicals

Lesson 5 · MA.8.NSO.1.7 · 7 questions

17. $\sqrt{169}$

13

18. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

19. $\sqrt{7^{2}-24}$

5

Finish under the bar: 49 − 24 = 25.

20. $\sqrt[3]{\frac{1}{27}}$

$\frac{1}{3}$

Top and bottom each get the root.

21. $\sqrt{100}\cdot4^{-2}$

$\frac{5}{8}$

$10\cdot\frac{1}{16}$.

22. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2+7}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

23. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

No. A cube root asks for the number used three times, and $(-4)(-4)(-4) = -64$ ***, so the value is −4. It is a square root of a negative number that is not a real number.***

This is the first misconception the state guide names for this benchmark.

PART E. The Order of Operations

Lesson 6 · MA.8.NSO.1.7 · 7 questions

24. $24 - 3 \cdot 5 + 1$

10

Multiply first, then left to right.

25. $48 \div 6 \cdot 2$

16

One level, left to right: 8 · 2. Not 4.

26. $4 + 3(8-5)^2$

31

$4 + 3(9)$.

27. $\frac{6^2+9}{3^2-4}$

9

45 over 5.

28. $-6^2 + (-6)^2$

0

The exponent attaches to its own base; the parentheses decide what that base is.

29. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

9

$20 \cdot \frac{9}{20}$.

30. Which expression is equivalent in value to $2+4\cdot5^2$?

- ○ A. 900
- ● **B.** 102
- ○ C. 42
- ○ D. 150

B

PART F. Real-World Problems and the Order of Operations *Lessons 7–8 · MA.8.NSO.1.7 · 5 questions*

31. Evaluate $5x^2$ when $x=-3$.

45

$5(-3)^2=5(9)$.

32. Kinetic energy is $K=\frac{1}{2}mv^2$. Find K when $m=400$ kg and $v=7$ m/s.

9,800 joules

$200(49)$ *. The unit is part of the answer.*

33. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

78 dollars and 364 dollars

$30+12(4)=78$ *and* $30(12)+4=364$ *. The same three numbers and the same words; what decides is which amount is repeated.*

34. SSDD. Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery. Same surface, four different questions: a measurement, an expression, an evaluation, and a size read back out of a total. Only part b has no number for an answer, and only part d runs the structure backwards.

a. What is the area of the patio?

225 square feet

b. Write an expression for the total cost.

$15^2(2.20)+60$

The rate multiplies the area; the flat fee does not.

c. Find the total cost.

555 dollars

$225(2.20)=495$ *, then* $+60$.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

20 feet

$940-60=880$ *, and* $880\div2.20=400$ *, whose square root is 20.*

35. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ○ A. $18^2(3.50 + 140)$
- ○ B. $18(3.50) + 140$
- ○ C. $18^2 + 3.50 + 140$
- ● **D.** $18^2(3.50) + 140$

D

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points MA.8.NSO.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

5.7×10^5

Same power: add the coefficients.

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

2.5×10^{-6}

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

1.43×10^9

14.3×10^8 *is the value; 14.3 is not less than 10, so it is not the answer.*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

4.57×10^4

$3.7 \times 10^3 = 0.37 \times 10^4$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

6×10^{10}

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

8×10^5

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

2×10^5

20×10^4 *is not finished.*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3×10^{-4}

0.3×10^{-3} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ● **C.** 2.914×10^5
- ○ D. 4.3×10^8

C

Scoring: 1 point.

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

A, C, D and F

C gives 30×10^5 *, which is* 3×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 6.123×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$2.3 \times 10^4 = 0.023 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1}. What was the number? Write your answer in scientific notation. *1 pt*

5×10^{-5}

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed.

$\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.5×10^{-4} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation

4 points MA.8.NSO.1.6

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

3.24×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 5.6×10^2 km in 8 hours. Find its average speed. *1 pt*

7×10^1 ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^3 mm and 4.1×10^2 mm. Find their sum. *1 pt*

6.66×10^3 ***mm***

Match the powers first: $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

8.828×10^{-2} ***g***

Match the powers first: $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

MA.8.NSO.1.7

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

−3

The power first, then the product, then the subtraction.

b. $\dfrac{5^2 + 7}{2^3}$ *1 pt*

4

32 over 8; the bar groups the whole numerator.

c. $36 \div 4 \cdot 3$ *1 pt*

27

One level, left to right: 9 · 3. Not 3.

d. $-4^2 + (-4)^2$ *1 pt*

0

−16 + 16. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

7

Finish under the bar: 36 + 13 = 49.

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5 + \frac{1}{4}$.

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9} + \frac{2}{3}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$14^2(2.40) + 85$

The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

555.40 dollars

196(2.40) = 470.40, then + 85.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

B

Scoring: 1 point.

13. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☒ **B.** $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☒ **D.** $\left(\frac{1}{6}\right)^{-1}$
- ☒ **E.** $\sqrt{5^2 + 11}$
- ☒ **F.** $\frac{4^2 + 8}{4}$

B, D, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken: $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.

Scoring: 1 point.

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

676 square inches

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.NSO.1.5	1, 2, 3, 4, 5, 6	12	
MA.8.NSO.1.6	7, 8	4	
MA.8.NSO.1.7	9, 10, 11, 12, 13, 14, 15	13	
Total		**29**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is

arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation

12 points MA.8.NSO.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^{6}) + (2.4 \times 10^{6})$ *1 pt*

6.7×10^{6}

Same power: add the coefficients.

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

5.5×10^{-5}

c. $(6.4 \times 10^{7}) + (8.2 \times 10^{7})$ *1 pt*

1.46×10^{8}

14.6×10^{7} *is the value; 14.6 is not less than 10, so it is not the answer.*

d. $(5.3 \times 10^{5}) + (2.6 \times 10^{4})$ *1 pt*

5.56×10^{5}

$2.6 \times 10^{4} = 0.26 \times 10^{5}$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

8×10^8

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

4×10^5

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

3×10^3

30×10^2 *is not finished.*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3×10^{-5}

0.3×10^{-4} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ● **B.** 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

B

Scoring: 1 point.

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

- ☒ **A.** $(2.5 \times 10^6) + (1.5 \times 10^6)$
- ☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
- ☒ **C.** $(8 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(4.3 \times 10^6) - (3 \times 10^5)$
- ☒ **E.** $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
- ☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

A, C, D and E

C gives 40×10^5 *, which is* 4×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 5.234×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$3.4 \times 10^4 = 0.034 \times 10^6$ *.*

Scoring: 1 point.

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

6×10^{-7}

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed.

$\frac{3 \times 10^{-2}}{5 \times 10^4} = 0.6 \times 10^{-6} = 6 \times 10^{-7}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.6×10^{-6} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation

4 points MA.8.NSO.1.6

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

4.375×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 3.5 · 1.25 = 4.375, and $10^2 \cdot 10^2 = 10^4$ *.*

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

9×10^1 ***km/h***

$\frac{7.2 \times 10^2}{8} = 0.9 \times 10^2$ *, which is not finished: 0.9 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

7.87×10^3 ***mm***

Match the powers first: $5.2 \times 10^2 = 0.52 \times 10^3$ *, and 7.35 + 0.52 = 7.87.*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

8.039×10^{-2} ***g***

Match the powers first: $3.61 \times 10^{-3} = 0.361 \times 10^{-2}$ *, and 8.4 − 0.361 = 8.039.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

MA.8.NSO.1.7

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

8

The power first, then the product, then the subtraction.

b. $\frac{4^2 + 9}{5}$ *1 pt*

5

25 over 5; the bar groups the whole numerator.

c. $45 \div 9 \cdot 3$ *1 pt*

15

One level, left to right: 5 · 3. Not 5.

d. $-3^2 - (-3)^2$ *1 pt*

−18

−9 − 9. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

9

Finish under the bar: 64 + 17 = 81.

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

$-\frac{35}{9}$

$-4 + \frac{1}{9}$.

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

1

$\frac{1}{4} + \frac{3}{4}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$12^2(2.60) + 75$

The rate multiplies the area; the flat fee does not. An expression that charges 75 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

449.40 dollars

144(2.60) = 374.40, then + 75.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ● **D. 39**

D

Scoring: 1 point.

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☒ **A.** $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☒ **C.** $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☒ **E.** $\frac{5^2+7}{4}$
- ☒ **F.** $\sqrt{6^2+28}$

A, C, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 36 + 64 is finished before the root is taken: $\sqrt{100} = 10$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '10' alone does not earn it, and neither does 'he added wrong'.

Scoring: 1 point.

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

900 square inches

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is $\sqrt{225} = 15$ *, so the second has side 30 and area* $30^2 = 900$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 450 has not gone through the side.*

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.NSO.1.5	1, 2, 3, 4, 5, 6	12	
MA.8.NSO.1.6	7, 8	4	
MA.8.NSO.1.7	9, 10, 11, 12, 13, 14, 15	13	
Total		**29**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is

arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.